

Unit 2: Computer Hardware

Teaching Point: Begin by defining "hardware" as the physical, tangible components of a computer system that you can touch and see, contrasting it with software.

1. Input Devices

- **Teaching Point:** Explain that input devices are used to enter raw data and instructions into the computer.
- **Keyboard:** The most common input device. Discuss its layout (QWERTY) and the different types of keys (alphanumeric, functional, control, and navigation keys).
- **Mouse:** A fundamental pointing device used to interact with graphical user interfaces (GUI). Cover basic actions like clicking, double-clicking, right-clicking, and dragging.
- **Scanner:** Used to convert physical documents, photos, or text into a digital format (soft copy) that the computer can store and process.
- **Barcode Reader:** An optical scanner that reads printed barcodes. *Example for students: Commonly used at supermarket checkout counters for billing and inventory management.*
- **MICR (Magnetic Ink Character Recognition):** A specialized technology used primarily by the banking industry to securely process and clear cheques by reading characters printed in special magnetic ink.
- **OCR (Optical Character Recognition):** Technology that recognizes text within a digital image (like a scanned document) and converts it into editable, searchable machine-encoded text.
- **OMR (Optical Mark Recognition):** A process of capturing human-marked data from document forms. *Example for students: Automatically checking multiple-choice exam answer sheets.*
- **Webcam:** A video camera that feeds or streams an image or video in real-time to or through a computer to a computer network (used for video conferencing).

2. Output Devices

- **Teaching Point:** Explain that output devices receive the processed data from the computer and translate it into a human-readable form (visual, audio, or physical).
- **Monitor:** The primary visual display unit. It provides a "soft copy" output. You might briefly mention different types, like LCD and LED screens.
- **Printer:** Produces a physical "hard copy" of digital documents or images on paper.
- **Plotter:** A specialized output device used to print high-quality, large-scale vector graphics. *Example for students: Used by engineers and architects for printing blueprints, maps, and architectural designs.*
- **Speaker:** Converts digital audio signals from the computer into sound waves, allowing users to hear music, alerts, and speech.

3. Storage Devices

- **Teaching Point:** Describe storage devices as the components where the computer permanently or temporarily holds data, applications, and the operating system.
- **Hard Disk (HDD):** The traditional magnetic storage device. It offers high-capacity storage for the operating system, software, and files but has moving mechanical parts.
- **SSD (Solid State Drive):** A newer, flash-based storage technology. Emphasize that unlike HDDs, SSDs have no moving parts, making them significantly faster, more durable, and quieter.
- **CD/DVD:** Optical storage media. CDs are typically used for audio and small amounts of data, while DVDs have a higher capacity suitable for video and larger software installations.
- **Pen Drive:** Also known as a USB flash drive. A small, highly portable storage device that plugs into a USB port, ideal for transferring files between computers.
- **Memory Card:** A tiny, flat flash drive used mostly in portable devices like digital cameras, smartphones, and tablets to store photos, videos, and apps.